



10 CFR 50.73

NMP1L3436
November 19, 2021

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Nine Mile Point Nuclear Station, Unit 1
Renewed Facility Operating License No. DPR-63
Docket No. 50-220

Subject: NMP1 Licensee Event Report 2021-002, Revision 0, *Isolation of both Emergency Condensers due to loss of UPS 162A*

In accordance with the reporting requirements contained in 10 CFR 50.73(a)(2)(v)(B), please find enclosed NMP1 Licensee Event Report (LER) 2021-002, Revision 0, Isolation of both Emergency Condensers due to loss of UPS 162A.

There are no regulatory commitments contained in this letter.

Should you have any questions regarding the information in this submittal, please contact Brandon Shultz, Site Regulatory Assurance Manager, at (315) 349-7012.

Respectfully,

Schuerman,
Adam G

Digitally signed by Schuerman,
Adam G
DN: cn=Schuerman, Adam G
Date: 2021.11.17 12:09:38 -05'00'

Adam Schuerman
Plant Manager, Nine Mile Point Nuclear Station
Exelon Generation Company, LLC

AS/JRA

Enclosure: NMP1 Licensee Event Report 2021-002, Revision 0, Isolation of both Emergency Condensers due to loss of UPS 162A

cc: NRC Regional Administrator, Region I
NRC Resident Inspector
NRC Project Manager

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NAR

Enclosure

NMP1 Licensee Event 2021-002, Revision 0
Isolation of both Emergency Condensers due to loss of UPS 162A
Nine Mile Point Nuclear Station, Unit 1

Renewed Facility Operating License No. DPR-63

NRC FORM 366 (08-2020)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104			EXPIRES: 08/31/2023											
LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)										Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: oir_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.									
1. Facility Name Nine Mile Point Unit 1					2. Docket Number 05000			3. Page 220 1 OF 5											
4. Title Isolation of both Emergency Condensers due to loss of UPS 162A																			
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved										
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number									
09	25	2021	2021	- 002 -	00	11	19	2021	N/A	05000 N/A									
									Facility Name	Docket Number									
									N/A	05000 N/A									
9. Operating Mode N: Power Operating Condition						10. Power Level 100%													
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)																			
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)											
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73											
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.69(g)		☒ 50.73(a)(2)(v)(B)		☐ 73.71(a)(4)											
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.71(a)(5)											
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(1)(i)											
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(i)											
☐ 20.2203(a)(2)(iii)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.77(a)(2)(ii)											
☐ 20.2203(a)(2)(iv)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)													
☐ 20.2203(a)(2)(v)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)													
☐ OTHER (Specify here, in abstract, or NRC 366A).																			
12. Licensee Contact for this LER																			
Licensee Contact Brandon Shultz, Site Regulatory Assurance Manager								Phone Number (Include area code) (315) 349-7012											
13. Complete One Line for each Component Failure Described in this Report																			
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS										
B	EF	UJX	Ametek	YES	N/A	N/A	N/A	N/A	N/A										
14. Supplemental Report Expected					15. Expected Submission Date			Month	Day	Year									
☒ No					☐ Yes (If yes, complete 15. Expected Submission Date)			NA	NA	NA									
16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)																			
On Saturday September 25, 2021, at approximately 0341 hours with power level at approximately 100 percent, Nine Mile Point Unit 1 (NMP1) experienced a loss of Uninterruptible Power Supply (UPS) 162A which resulted in the loss of Reactor Protection System (RPS) Bus 11. The loss of RPS Bus #11 resulted in a half scram, Reactor Water Cleanup isolation, and isolation of both #11 and #12 Emergency Condensers. The cause of this event was the premature failure of the oscillator board (X2) and/or the sync board (X5) in UPS 162A. Corrective actions include replacement of faulted sync and oscillator boards. This event is reportable under 10CFR50.73(a)(2)(v)(B).																			



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		2. DOCKET NUMBER		3. LER NUMBER		
Nine Mile Point Unit 1		05000-		YEAR	SEQUENTIAL NUMBER	REV NO.
		220		2021	002	00

NARRATIVE

I. DESCRIPTION OF EVENT

A. PRE-EVENT PLANT CONDITIONS:

Unit 1 was in Run, operating at approximately 100% rated thermal power when the event occurred. There were no structures, systems, or components out of service that contributed to this event.

B. EVENT:

On September 25, 2021, at approximately 0341 hours, the NMP Unit 1 control room received multiple annunciators concurrent with a loss and subsequent restoration of RPS Bus #11 on UPS 162A. This resulted in a half scram, Reactor Water Cleanup isolation, and isolation of both #11 and #12 Emergency Condensers. The Emergency Condenser System was declared inoperable, and the appropriate Technical Specification Action was entered.

The condition was reported to the NRC on September 25, 2021 at approximately 0916 hours pursuant to the requirements of 10 CFR 50.72(b)(3)(v)(B) (Event Notification #55492).

After consultation with the vendor Ametek, the oscillator board (X2) and the sync board (X5) were replaced simultaneously. This was done based on advisement from Ametek that these cards should be replaced in pairs. Following replacement of these circuit boards, UPS-162A was restarted and exhibited no signs of frequency or voltage oscillations. Additionally, the UPS-162A was energized and the output monitored for several days to ensure consistent output performance while unloaded with no anomalies noted.

Nine Mile Point Unit 2 (NMP2) was unaffected.

The event has been entered into the plant's corrective action program as IR 04448532.

C. INOPERABLE STRUCTURES, COMPONENTS, OR SYSTEMS THAT CONTRIBUTED TO THE EVENT:

No other systems, structures, or components contributed to this event.



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Nine Mile Point Unit 1	05000- 220	YEAR 2021	SEQUENTIAL NUMBER 002	REV NO. 00

NARRATIVE

D. DATES AND APPROXIMATE TIMES OF MAJOR OCCURRENCES AND OPERATOR ACTIONS:

The dates, times, major occurrences, and operator actions for this event are as follows.

September 25, 2021:

0341 Received multiple Control Room alarms due to loss and subsequent restoration of RPS Bus #11 on UPS 162A. Both Emergency Condensers #11 and #12 isolate. Reactor Water Cleanup isolate.

0403 RPS Bus #11 de-energized by operators to prepare for dead bus transfer to I&C Bus 130A.

0405 RPS Bus #11 power was restored by operators via I&C Bus 130A

0420 Restored #11 Emergency Condenser to standby

0429 Restored #12 Emergency Condenser to standby

1031 RPS Bus #11 de-energized by operators to prepare for dead bus transfer to UPS 162B

1033 RPS Bus #11 power was restored by operators via UPS 162B

Both Emergency Condensers were inoperable for a total of 48 minutes from event initiation to restoration.

During the time Emergency Condensers were isolated, NMP1 Limiting Condition of Operation (LCO) 3.1.3.a was not met. Subsequently, the station entered LCO 3.1.3.e but normal orderly shutdown was not initiated since the first condenser restored in approximately 39 minutes, driving the station to LCO 3.1.3.b. Full compliance with LCO 3.1.3 was restored at 0429.

E. METHOD OF DISCOVERY:

This event was self-revealing as identified by multiple Control Room alarms.

F. SAFETY SYSTEM RESPONSES:

Upon loss of the RPS #11 Instrument Bus, the Emergency Condensers and Reactor Water Cleanup isolated as expected due to the voltage transient. Also, the momentary loss of power resulted in Reactor Building Emergency Ventilation initiation.



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Nine Mile Point Unit 1	05000-	YEAR	SEQUENTIAL NUMBER	REV NO.
		2021	002	00

NARRATIVE

II. CAUSE OF EVENT

The cause of this event was determined to be the premature failure of the UPS 162A oscillator board (X2) and/or the sync board (X5).

III. ANALYSIS OF THE EVENT

There were no actual nuclear safety consequences associated with this event. No initiating transient was present. Although the Emergency Condenser loops were automatically isolated, the conditions requiring automatic initiation of the system were not present. The event was caused by an equipment failure creating a voltage transient that led to the spurious isolation.

Although the Emergency Condensers isolated, the isolation signals are designed to be bypassed should they be required to operate. The ability to manually initiate the system is recognized in the Probabilistic Risk Assessment model supported by procedural guidance that directs restoration. It is concluded that the safety significance of this event is low and the event did not pose a threat to the health and safety of the public or plant personnel.

Isolation of both the Emergency Condensers is reportable under 10 CFR 50.72(b)(3)(v)(B) and 10 CFR 50.73(a)(2)(v)(B) as any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the removal of residual heat.

IV. CORRECTIVE ACTIONS

A. ACTION TAKEN TO RETURN AFFECTED SYSTEMS TO PRE-EVENT NORMAL STATUS:

Operators took immediate actions and transferred the RPS Bus #11 to I&C Bus 130A. The Emergency Condensers #11 and #12 were returned to full standby service September 25, 2021 at 0420 and 0429 respectively. Subsequently, RPS Bus #11 was then restored to power by operators via UPS 162B.

B. ACTION TAKEN OR PLANNED TO PREVENT RECURRENCE:

The faulted sync and oscillator boards were replaced.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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NARRATIVE**V. ADDITIONAL INFORMATION****A. FAILED COMPONENTS:**

Uninterruptible Power Supply

B. PREVIOUS LERs ON SIMILAR EVENTS:

NMP LER 2016-002-00 reported the Isolation of both Emergency Condensers due to loss of UPS 162B. The cause of the LER 2016-002-00 were determined to be the failure of a UPS capacitor and the bypass power transfer set point being set too low for the type of transient.

C. THE ENERGY INDUSTRY IDENTIFICATION SYSTEM (EII) COMPONENT FUNCTION IDENTIFIER AND SYSTEM NAME OF EACH COMPONENT OR SYSTEM REFERRED TO IN THIS LER:

COMPONENT	IEEE 803 FUNCTION IDENTIFIER	IEEE 805 SYSTEM IDENTIFICATION
Uninterruptible Power Supply	UJX	EF
Reactor Protection System	N/A	JC
Emergency Condenser System	BN	BL

D. SPECIAL COMMENTS

None